Advance Packaging in Energy Efficient World

ASE Group

September, 2024

Outline

- Problem statement
- VRM / IVR Power Delivery
- Photonics

ASE Confidential / Security-C

Why we need advanced efficiency solutions for power, for bandwidth

As Global wireless move to 5G, and the evolution of AI in data center, our industry is at an inflection point – where energy efficiency is the single most important driver of future improvement in compute.

The creation of ChatGPT, high compute power for NVIDA GPU-base boosting some forecast that predict data center could draw up to 21% of the world's electricity supply by 20230.

Nvidia's A100 max power consumption is 400W, and the H100's power consumption is 75% higher versus the A100 at 700W, Blackwell maximum power consumption is at 1200W. **The 300% increase in GPU power consumption just happen in less than four years.**

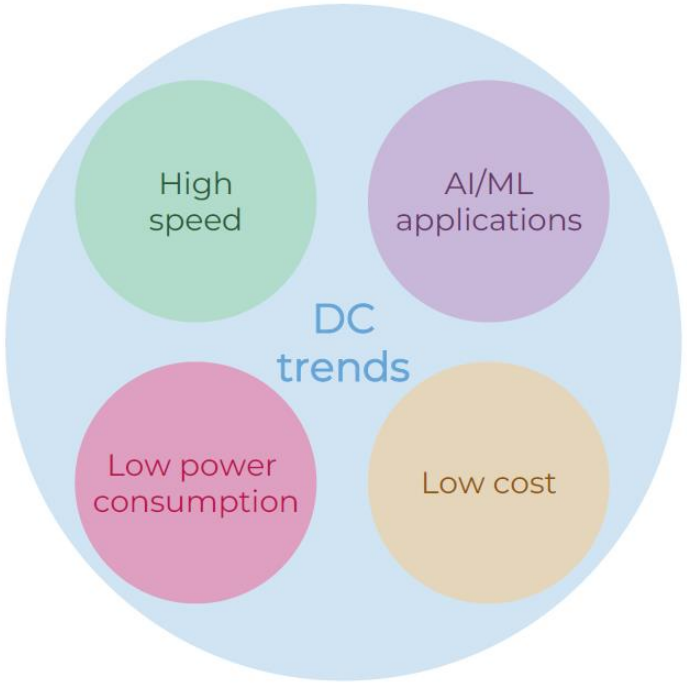
Silicon process and package technology advancement , improving efficiency of memory delivery, efficiency of IO remain essential.

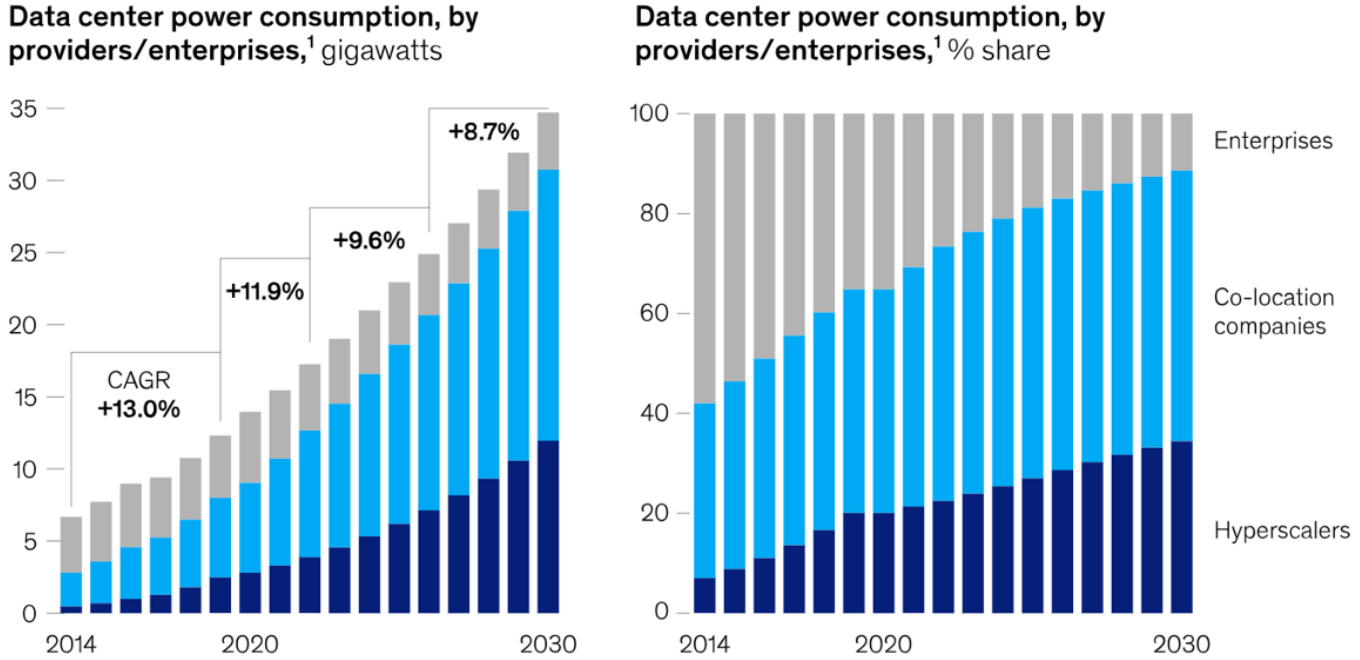
* Source: Forbes, Semiconductor Engineering

Problem statement

- Why we need advanced efficiency solutions for power, for bandwidth

- AI Training model parameters are skyrocketing & data center demand is forecast to grow by some 10 percent a year until 2030





Source: https://www.mckinsey.com/industries/technology-media-and-telecommunications/our-insights/investing-in-the-rising-data-center-economy

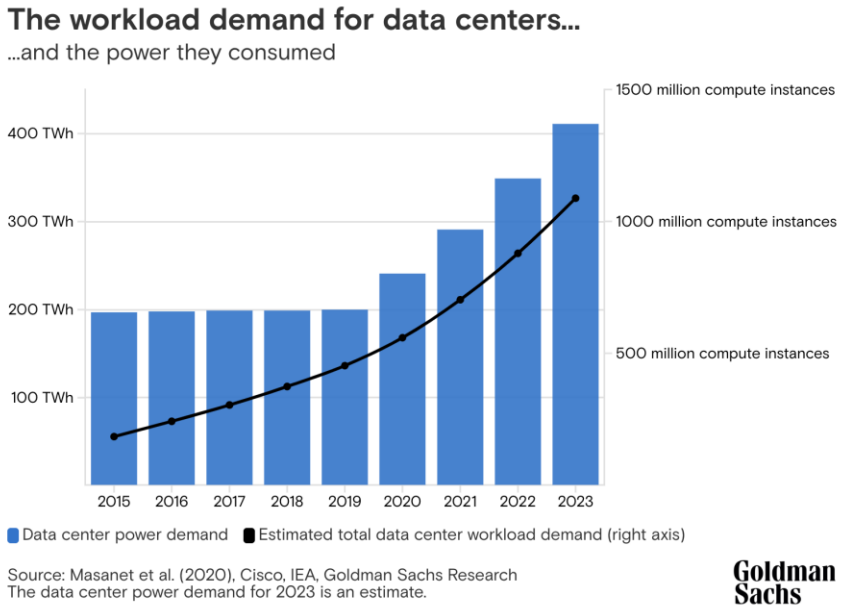
LLAMA 2 · Claude 3 · OpenAI ChatGPT 3.5 · OpenAI ChatGPT 4.0 · Gemini

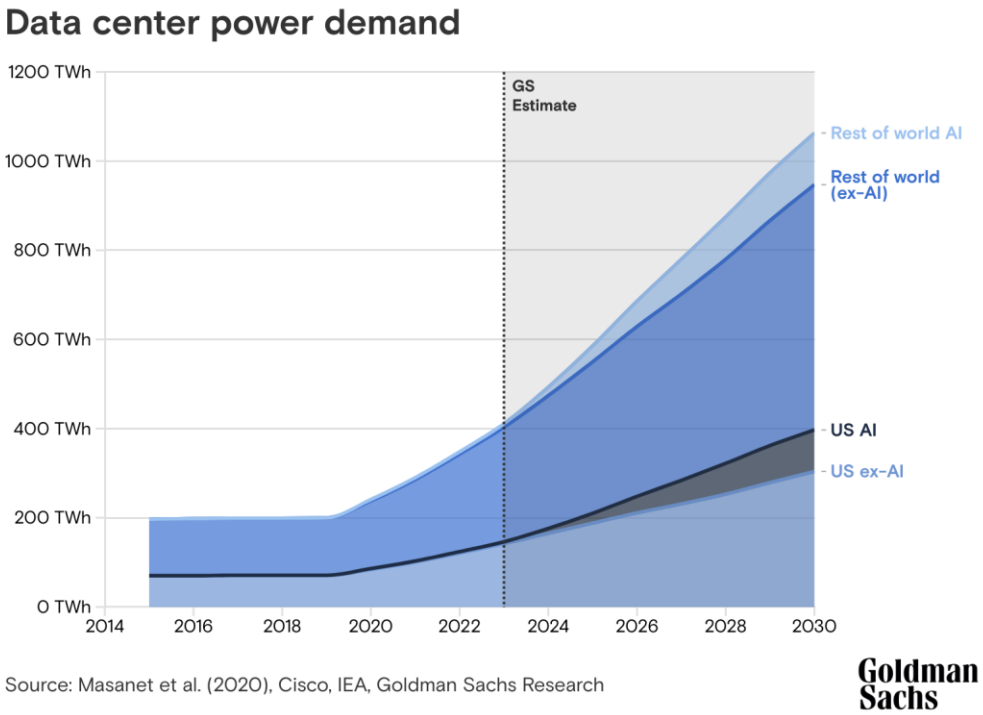
	Meta LLAMA	Amazon Claude	OpenAI GPT 3.5	OpenAI GPT 4	Google Gemini
Model Parameters	70B	130B	175B	1760B	1600B
Model Type	Pure text	Pure text	Pure text	Add picture recognition + text to picture	Add video/language comprehension
Other Features	• Open source, free • Easy for reuse • Vibrant user	• Charge API • Emphasis AI safety	• Charge API • Most widely used	• Charge API • Strong logic inference • Support long text length • Understand tables	• Charge API • Multi mode support: video, translation, etc.

Problem Statement - The Need of Advanced Efficiency

Tremendous Growth of AI and Data Center Power Demands

- **Pre-2020, Data center workload**s nearly tripled between 2015 and 2019 and data centers' demand for power remained flattish, at about 200 terawatt-hours per year.
- **Post-2020**, the power consumed by data centers has risen driven by AI boosting. A single ChatGPT query requires 2.9 watt-hours of electricity, compared with 0.3 watt-hours for a Google search.
- Goldman Sachs Research estimates the overall increase in data center power consumption from AI to be on the order of 200 terawatt-hours per year between 2023 and 2030. By 2028, they expect AI to represent about 19% of data center power demand.



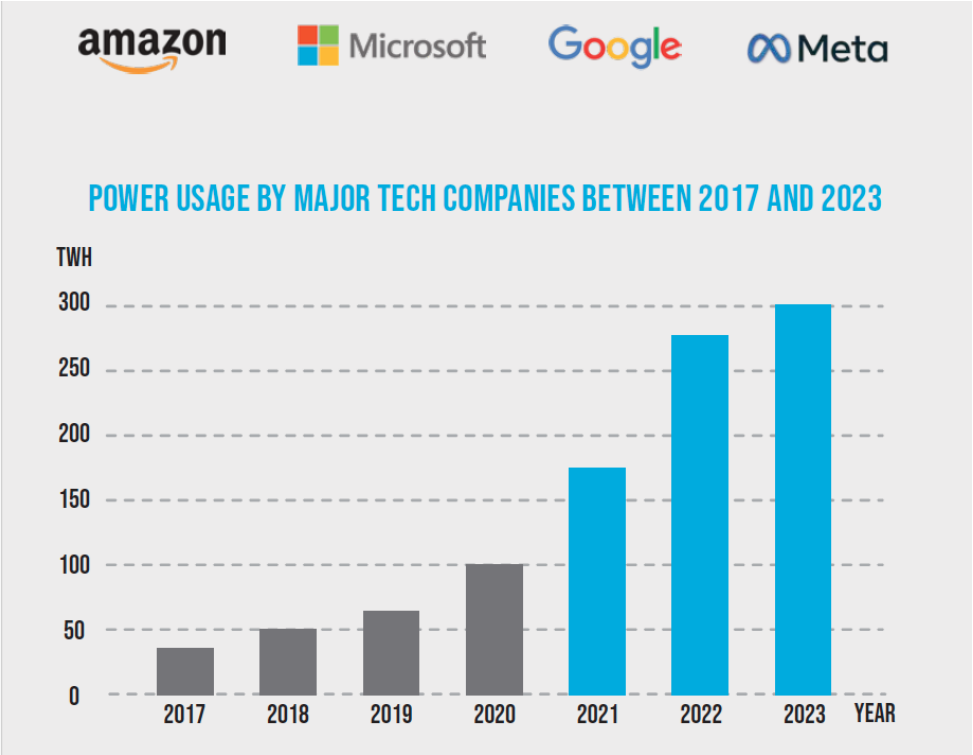


https://www.goldmansachs.com/insights/articles/AI-poised-to-drive-160-increase-in-power-demand

Problem Statement - The Need of Advanced Efficiency

High performance computing processors requires high power delivery

- The power used by Amazon, Microsoft, Google, and Meta, some of the largest owners and operators of data centers, has more than doubled from 2017 to 2021, using roughly 72 TWh combined in 2021. This equates to enough electricity to power roughly 6.72 million US households in 2021.

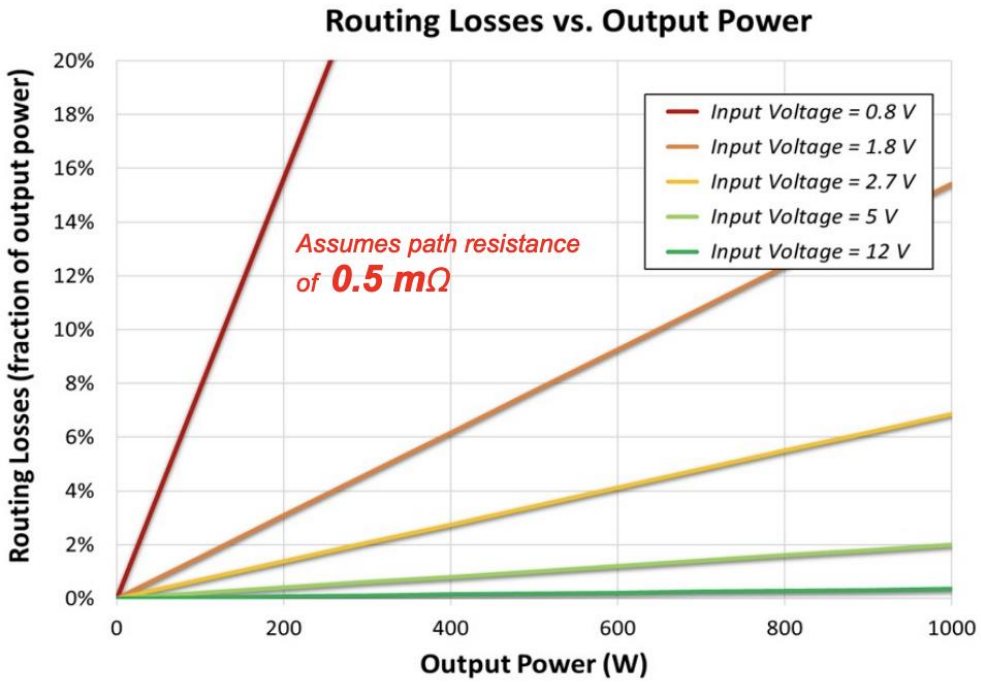


Source: IEA: https://www.iea.org/energy-system/buildings/data-centres-and-data-transmission-networks

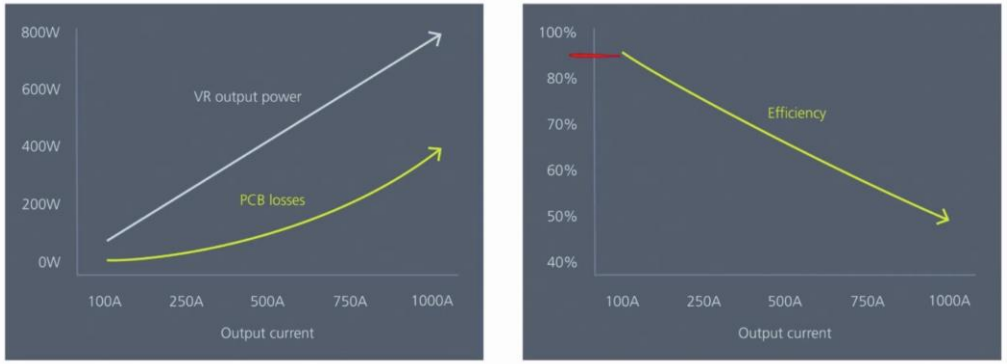
Problem Statement - The Need of Efficient Power Delivery

Vertical Power delivery

- High power at lower voltages requires high current and causes high power delivery loss (I^2R)
  - Power delivery need to be in short distance—Vertical power delivery
- Need to integrate effective power conversion from high voltage at point of load (PoL)
  - Integrate new power delivery architectures—Vertical power delivery



Source: 2024 IEEE VLSI Symposium on Technology & Circuit



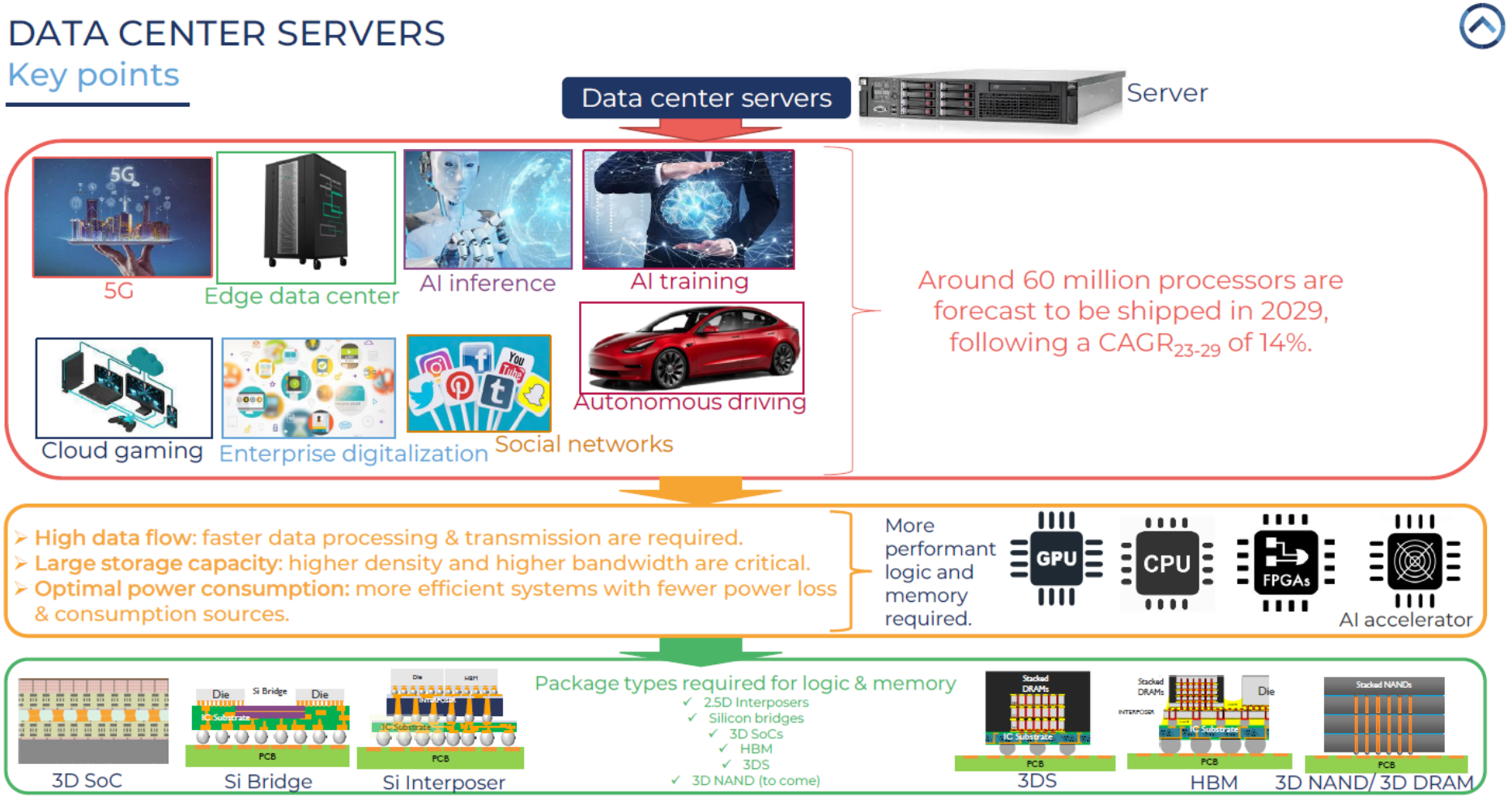
Example with PCB resistance of 400µΩ (VR at 0.8V_{OUT})

	Vicor Lateral	Conventional
PDN resistance	50µΩ	400µΩ
PDN loss @ 500 Amps	12.5W loss 96.8% efficiency	100W 75% efficiency
PDN loss @ 1000 Amps	50W loss 93.75% efficiency	400W 50% efficiency

PDN Power Loss, due to circuit board copper resistance = I^2R

Tremendous growth in AI power requirement – in blade, rack and datacenter

AI will cause data centers to consume even more energy.
The Electric Power Research Institute forecasts that data centers may see their electricity consumption more than double by 2030, reaching 9% of total electricity demand in the US. The IEA is projecting global electricity demand from AI, data centers and crypto to rise to 800 TWh in 2026 in its base case scenario, a nearly 75% increase from 460 TWh in 2022.



* Source: YOLE, Forbes

Bandwidth needs as compute power accelerate

Over the past decade, the tech industry has come to realize that the graphics processing units (GPUs) that power these gaming animations are also a good fit for data center applications like artificial intelligence (AI) and for modeling and visualization in high-performance computing (HPC). This is because the mathematical and graphical calculations offloaded to the GPU in those workloads are similar to the GPU functions in advanced gaming.

And what do these gaming, AI and HPC applications all have in common? They rely heavily on memory performance. The memory bandwidth in GPU determines how fast it can transfer data to and from between processing cores and memory.

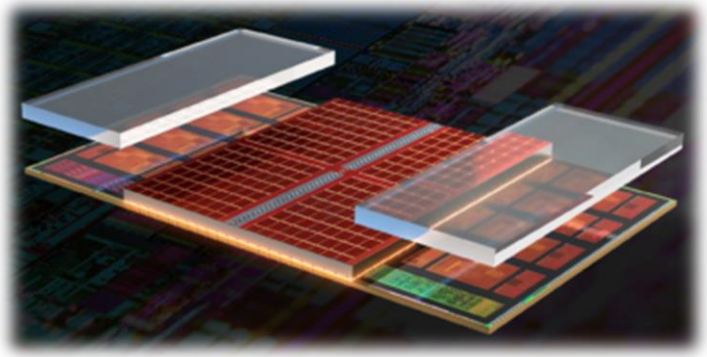
High-bandwidth memory offers faster data transfer rates using less power. With a smaller form factor, high-bandwidth memory is the go-to memory choice for today's data centers and tomorrow's AI and server solutions.

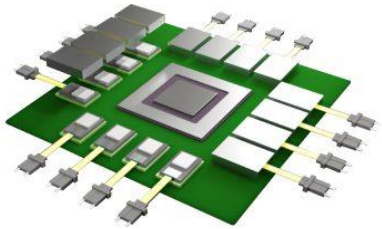
Table 3.4. HBM Market Projections

	2023	2024	2025	2026	2027
Number of HBM stacks (millions)	30	47	63	98	144
Number of DRAMs (millions)	250	390	663	1,080	1,691
Thousands of 300mm wafers (thousands)	544	927	1,655	2,831	4,654

* Source: Micron, TechSearch International Inc.

Advance Packaging Drivers Holistic System Level Efficiency





Silicon and Si Stacking

Application

Packaging

Optical Interconnect

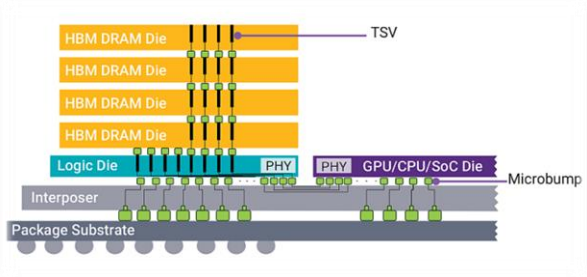
Compute

Energy efficiency through holistic design

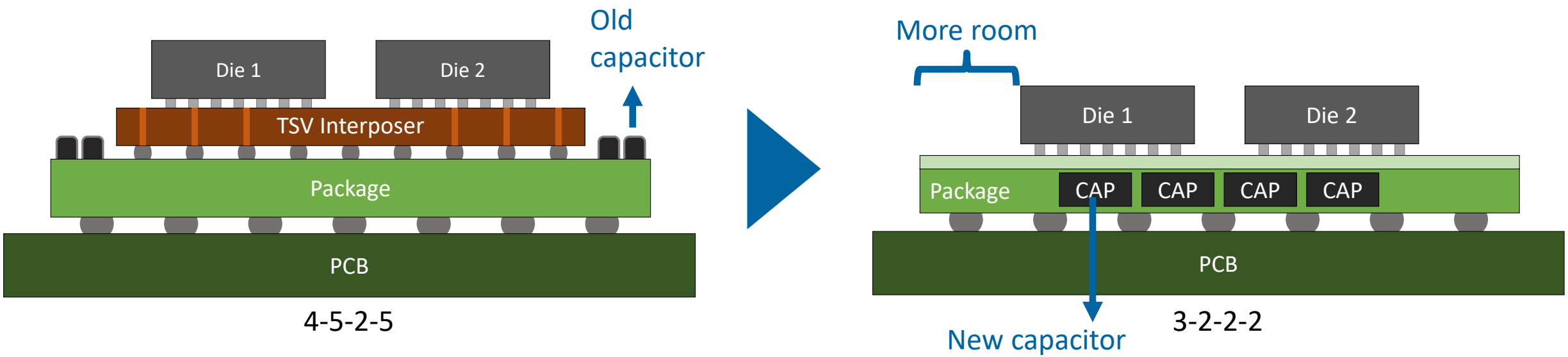
Software Stack

Memory

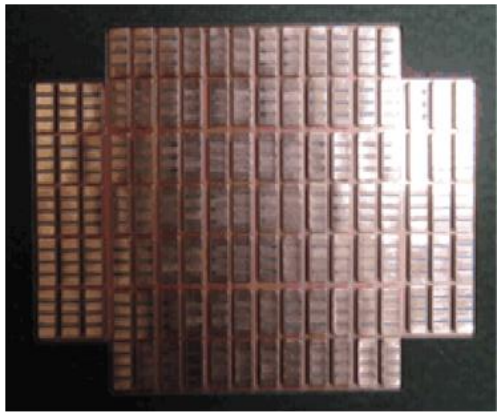
Accelerators



Integrating VRM to Package Substrate

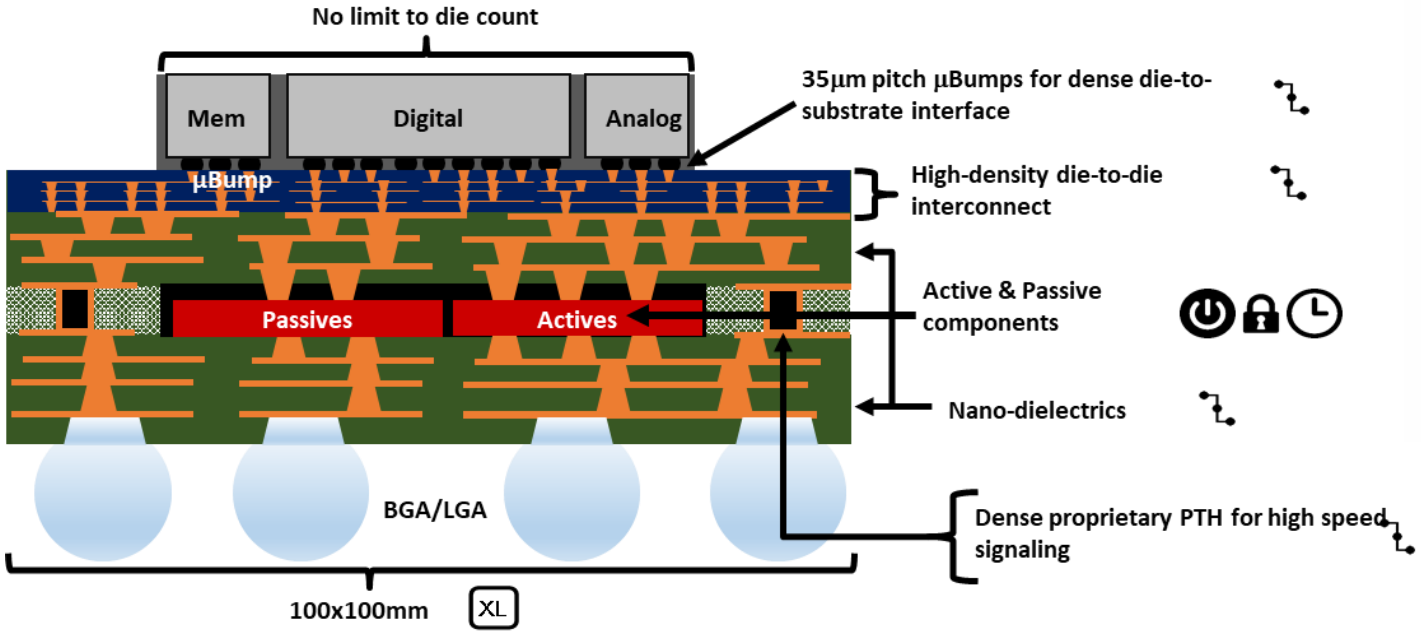


- Integrate high-density interconnect and decoupling caps
- Improves power delivery and IO speed while reducing metal layer count

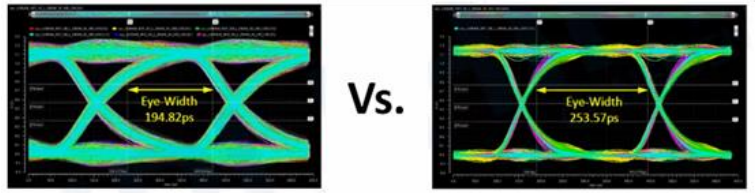


Voltage Regulator Module Deliver Power to the Chips

- PIM's cost & performance are better than after 7nm advanced node develop
- Decoupling frequency 100MHz ➔ 200MHz:
  - Clock & PHY VDD I/O 32% noise reduction
  - Driver VDD 78% noise reductio
- Voltage drop 179mV ➔ 119mV, 60mV improve could increase ~40% frequency
- Fast die-to-die interconnect and lower power via 30% wider eye

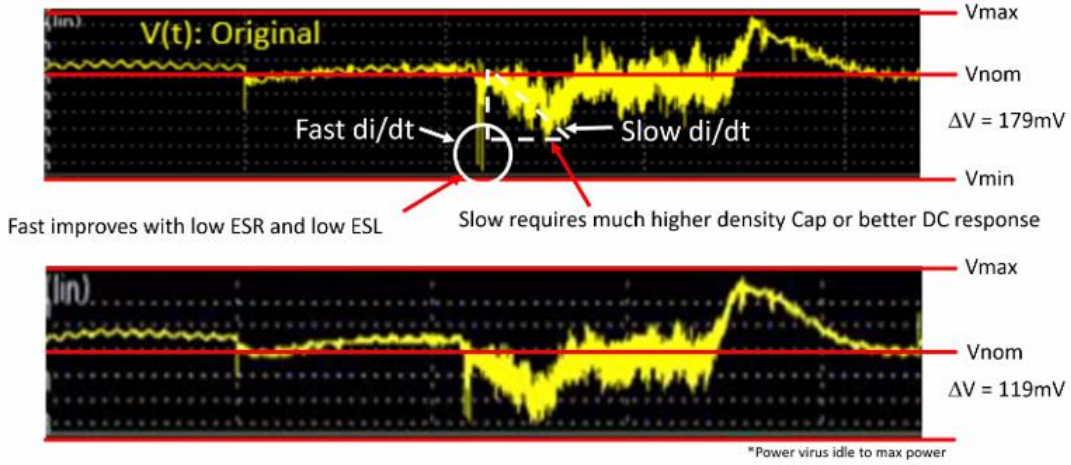


Better HBM Signaling

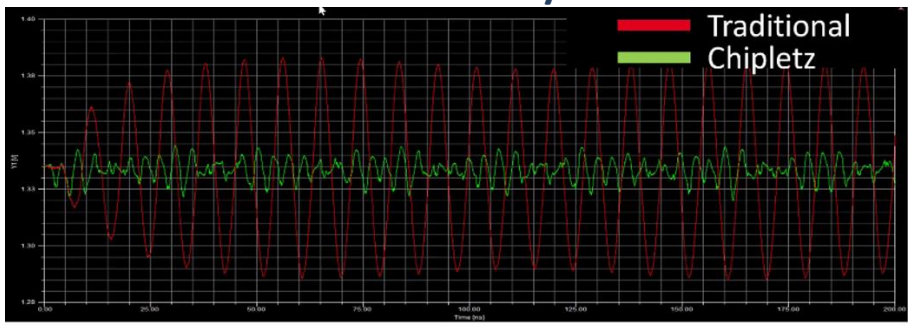


Silicon interposer | Chipletz substrate

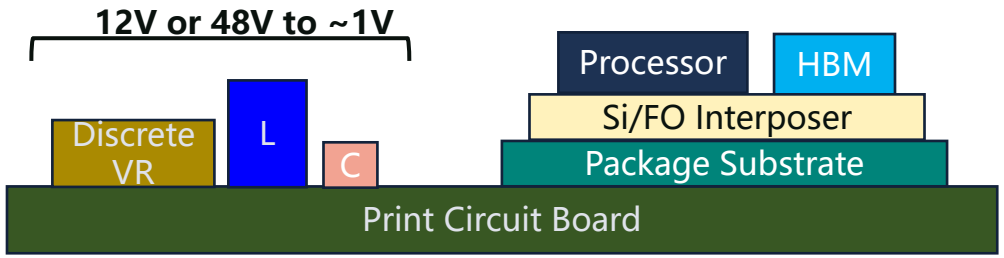
32% noise reduction in Vcore



78% noise reduction in I/O driver Vdd

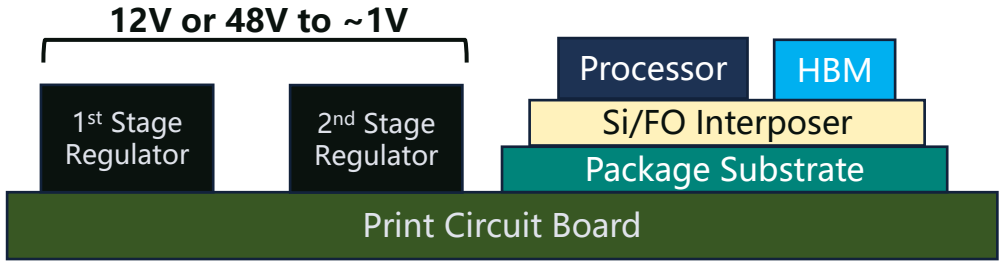


Power Solutions in HPC System

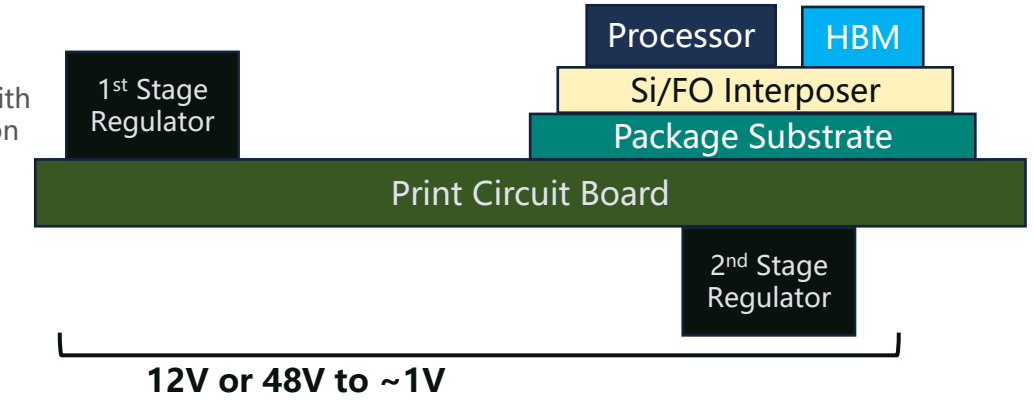


Typical systems using single stage 12V to ~1V

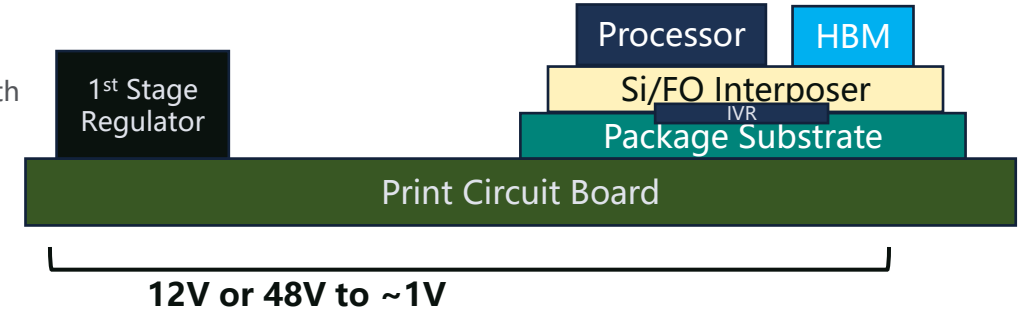
Side-by-Side Integration

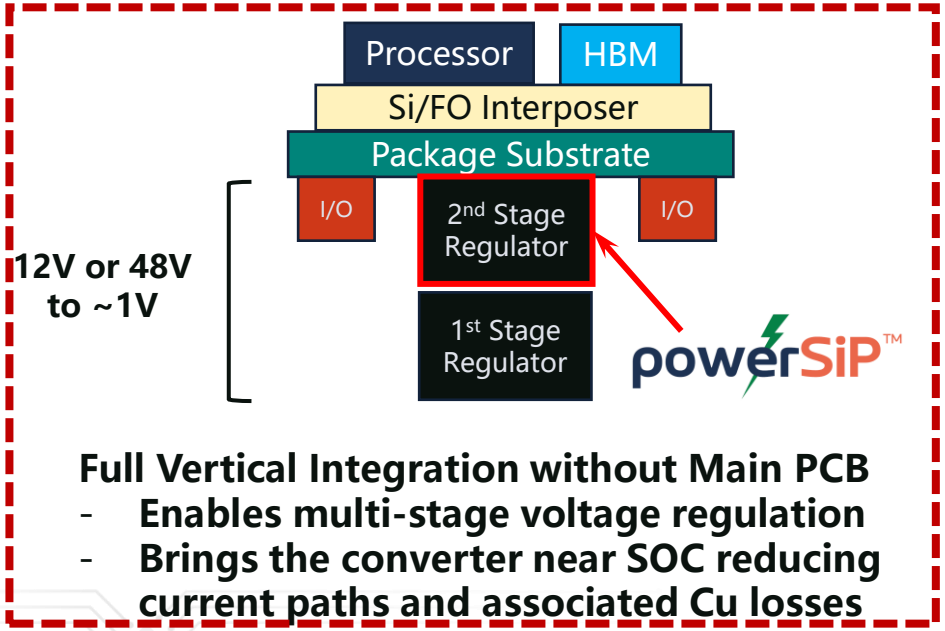


2nd Stage Power with Vertical Integration



2nd Stage Power with IVR Integration





Full Vertical Integration without Main PCB

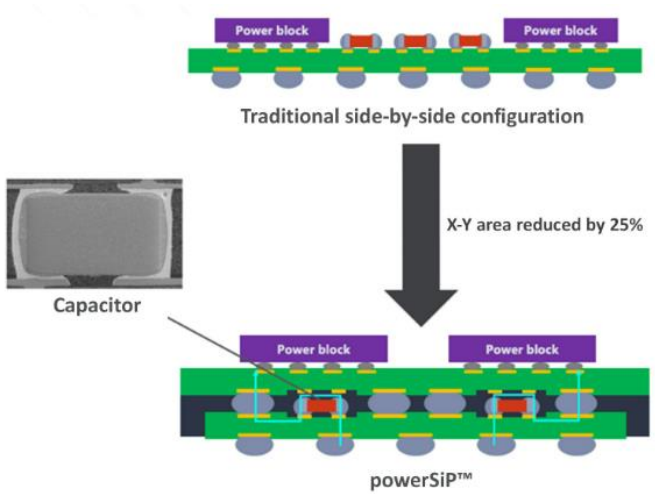
- **Enables multi-stage voltage regulation**
- **Brings the converter near SOC reducing current paths and associated Cu losses**

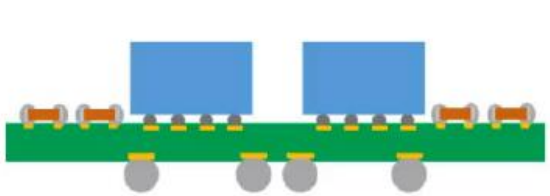
powerSiP™

VRM / IVR (powerSiP)

- Structure explanation / how it works

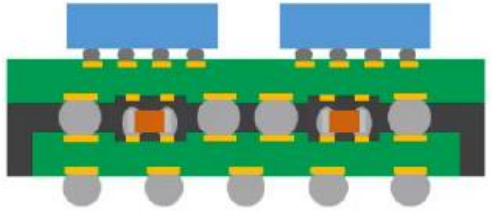
- powerSiP™ technology:
  - enables a vertically integrated multi-stage voltage regulator module (VRM) for higher system efficiency and lower power consumption
  - increases power efficiencies by 50% in AI and data center applications through increasing current density from 0.4 A/mm² to 0.6 A/ mm²
  - provides routing power loss reduction from 12% to 6%, signifying a 50% reduction over a side-by-side configuration
  - provides an option to place the voltage regulator directly under the SoC and chiplets, and vertical integration allows a large current supply at a short power delivery path
  - reduces impedance in the PDN thus improving system performance and functionality while simultaneously increasing overall efficiency and power density
  - designed to meet related data center requirements, performance expectations, and power improvements for AI and HPC applications





Total power = 720 A (120 A x 6)
Cap No.: 800 ea
Total area = 1190 + 480 = 1670 mm2

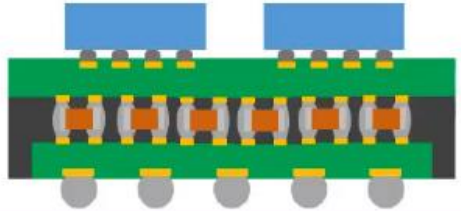
Power Density = 720/ 1670 = **0.4 A/mm2**



Total power = 720 A (120 A x 6)
Cap no: 800 ea
Total area = 1190 mm2

Gen 1

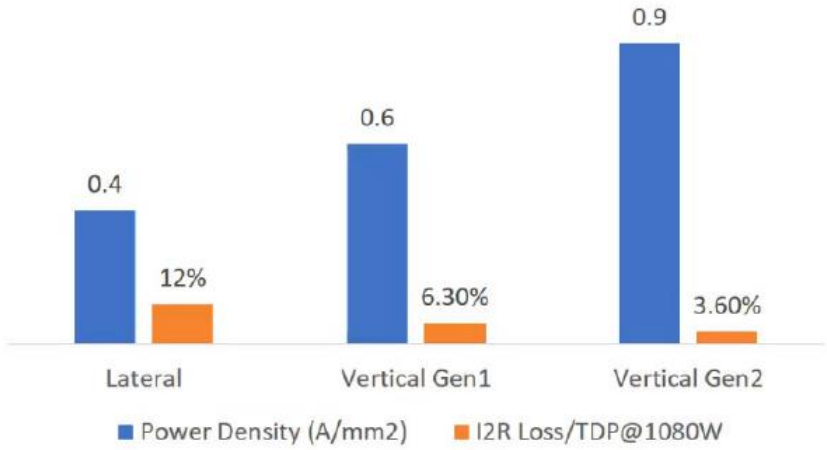
Power Density = 720/ 1190 = **0.6 A/mm2**



Total power = 1080 A (180 A x 6)
Cap no.: 1800 ea
Total area = 1190 mm2

Gen 2

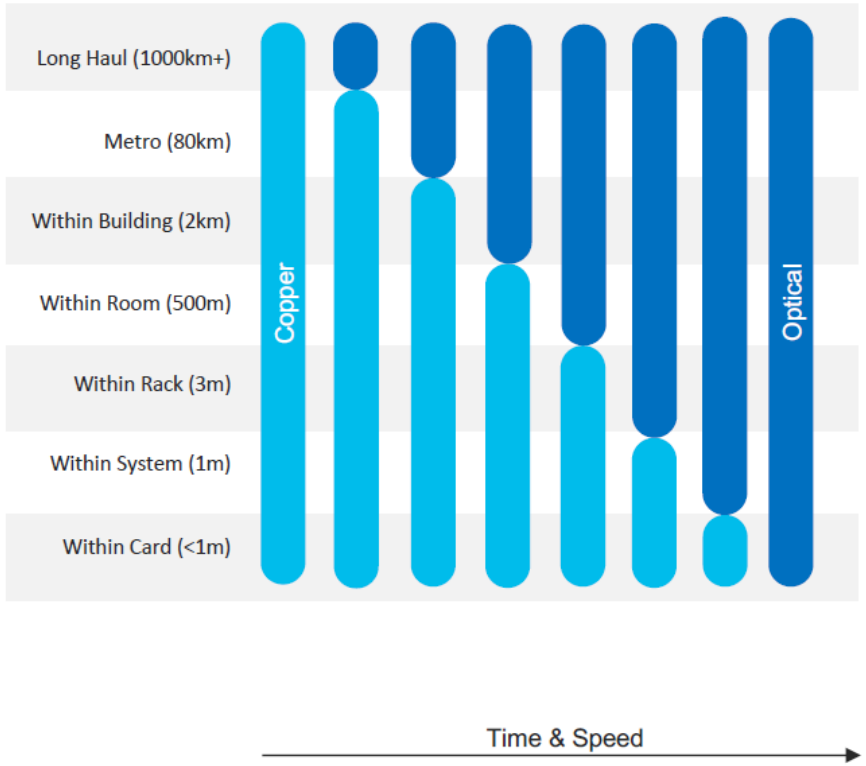
Power Density = 1080/ 1190 = **0.9 A/mm2**



Future System Capacity will only be possible with Silicon & Optical Integration

Si SILICON + Op OPTICS

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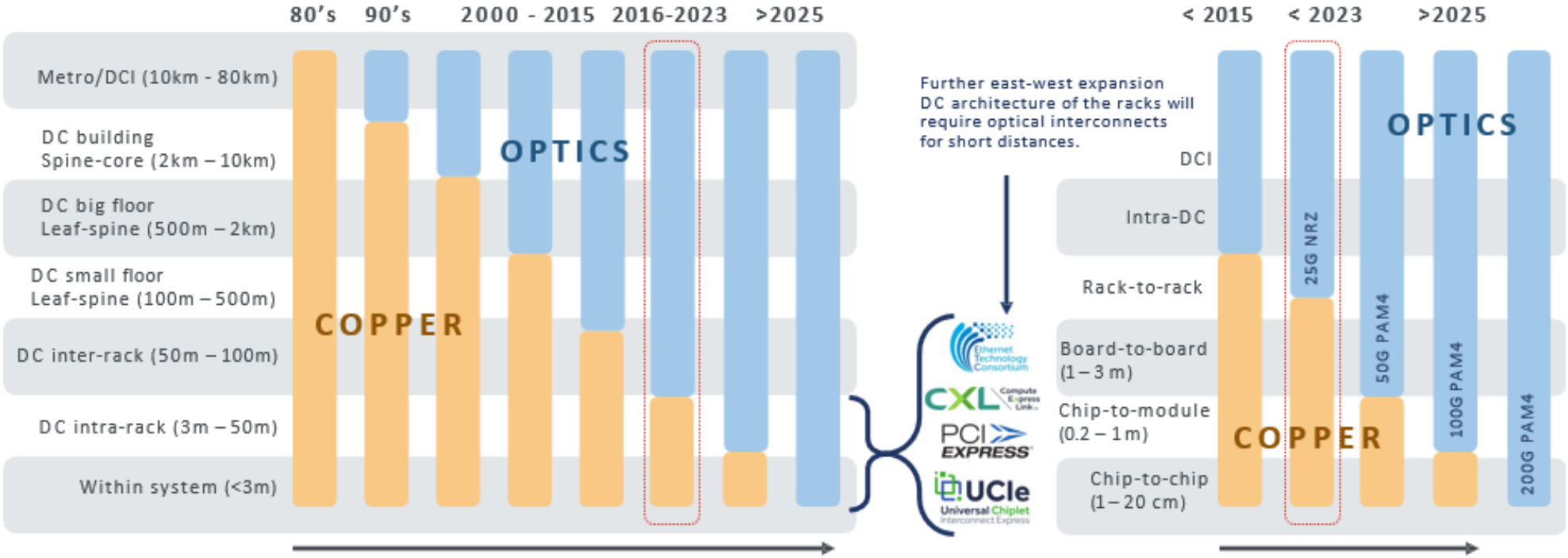
Copper to Optics

Optics evolution: moving from copper to optics

TREND:

Optical interconnects tend to penetrate rack and computing systems.

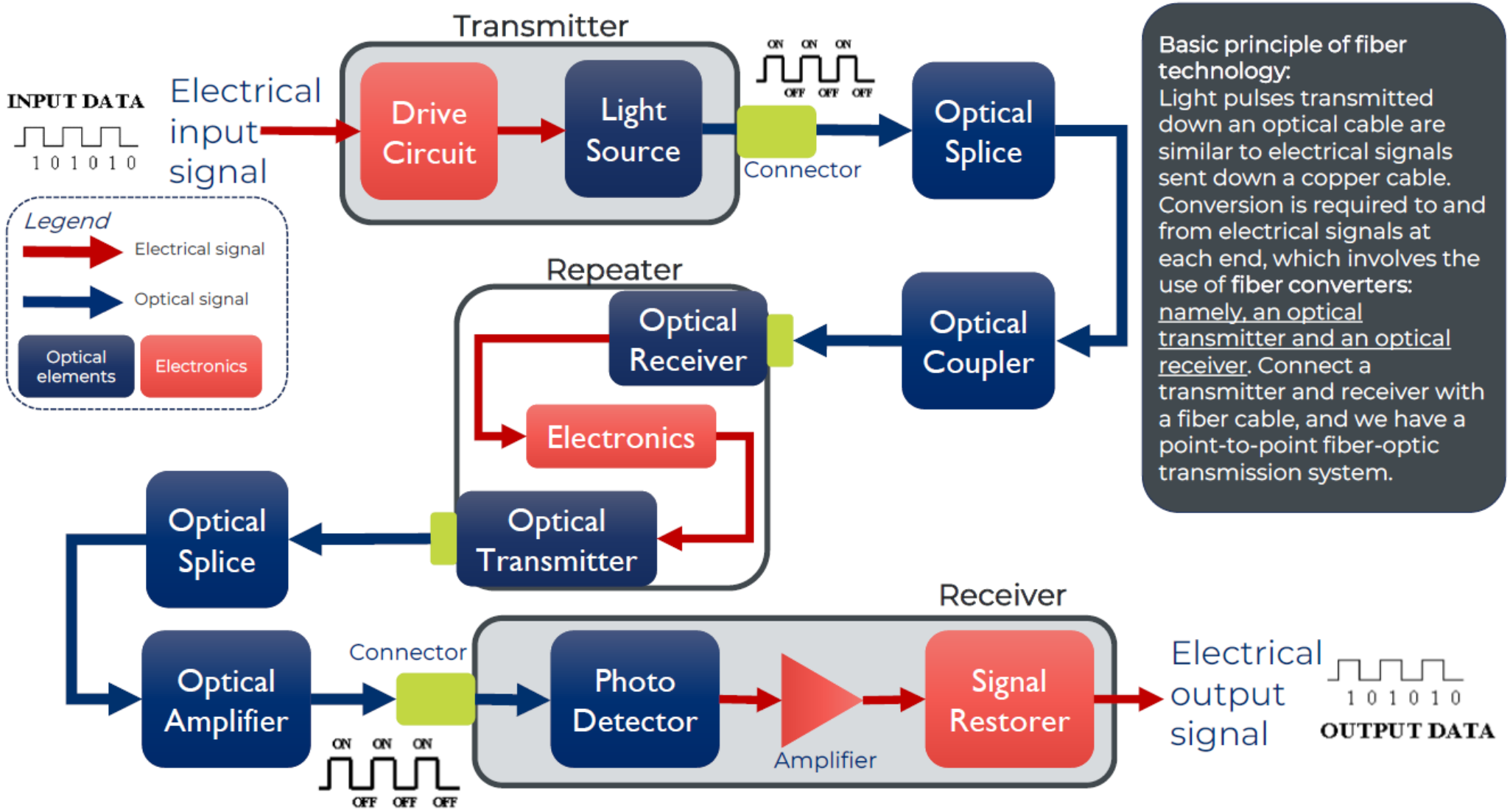
- In current networks, most links within the rack are copper.
- As SerDes rates increase, the longest copper links need to move to optical.
  - For 100 Gb/s per lane, the targeted reach is 2 m (IEEE 802.3ck).
- A TOR switch could be equipped with a mix of pluggable ports for its downlinks, and CPO ports for its uplinks.
- The DAC option remains attractive for server connectivity or cabled backplanes.
- Eventually, all the links leaving a silicon package will be optical rather than electrical.



Photonics

- Structure explanation / how it works

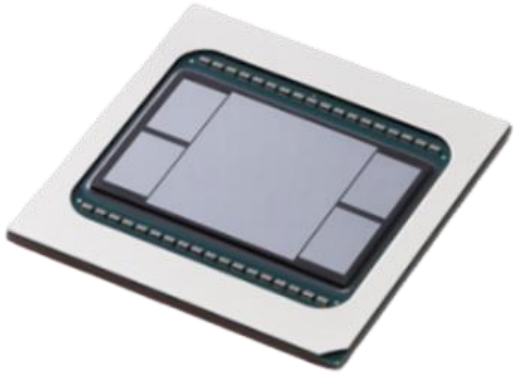
➢ Optical transceiver: basic principle



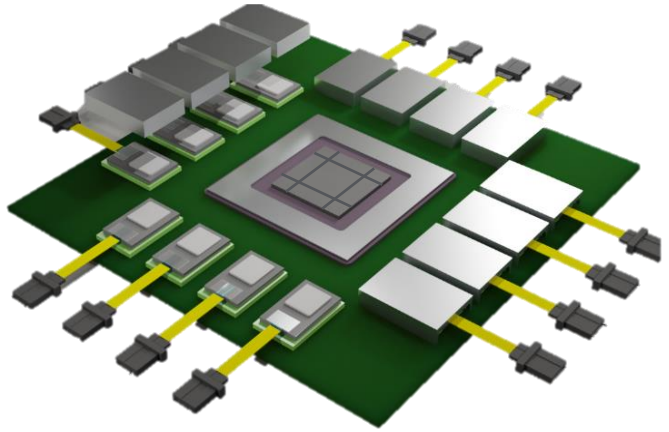
YOLE YINTR24404 - Optical Transceivers for Datacom and Telecom 2024

Si Photonics
has Similar Heterogeneous Integration

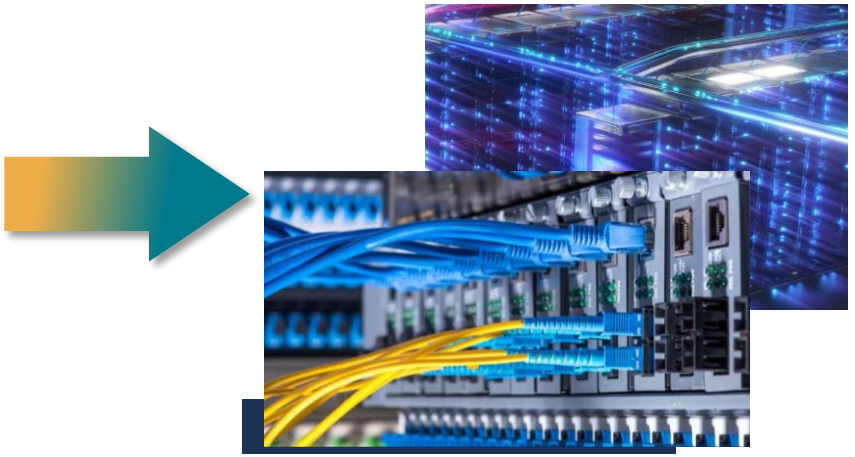
Chiplets



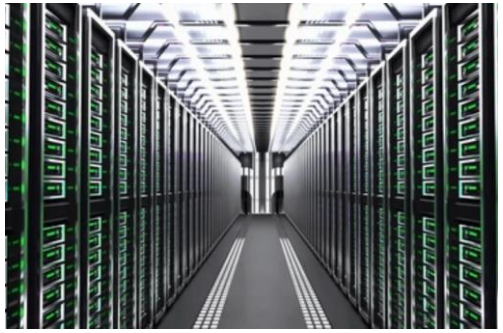
CPO (Co-Packaged Optics)



Data Center



Data Center Eco-System and Supplier Chain Synergy



Data Center Architect

System Requirement



CM
System Integration

New Generation ODM-OEM

Chip Design
PIC/EIC

Foundry
Chip MFG

OSAT
EIC/PIC Integration

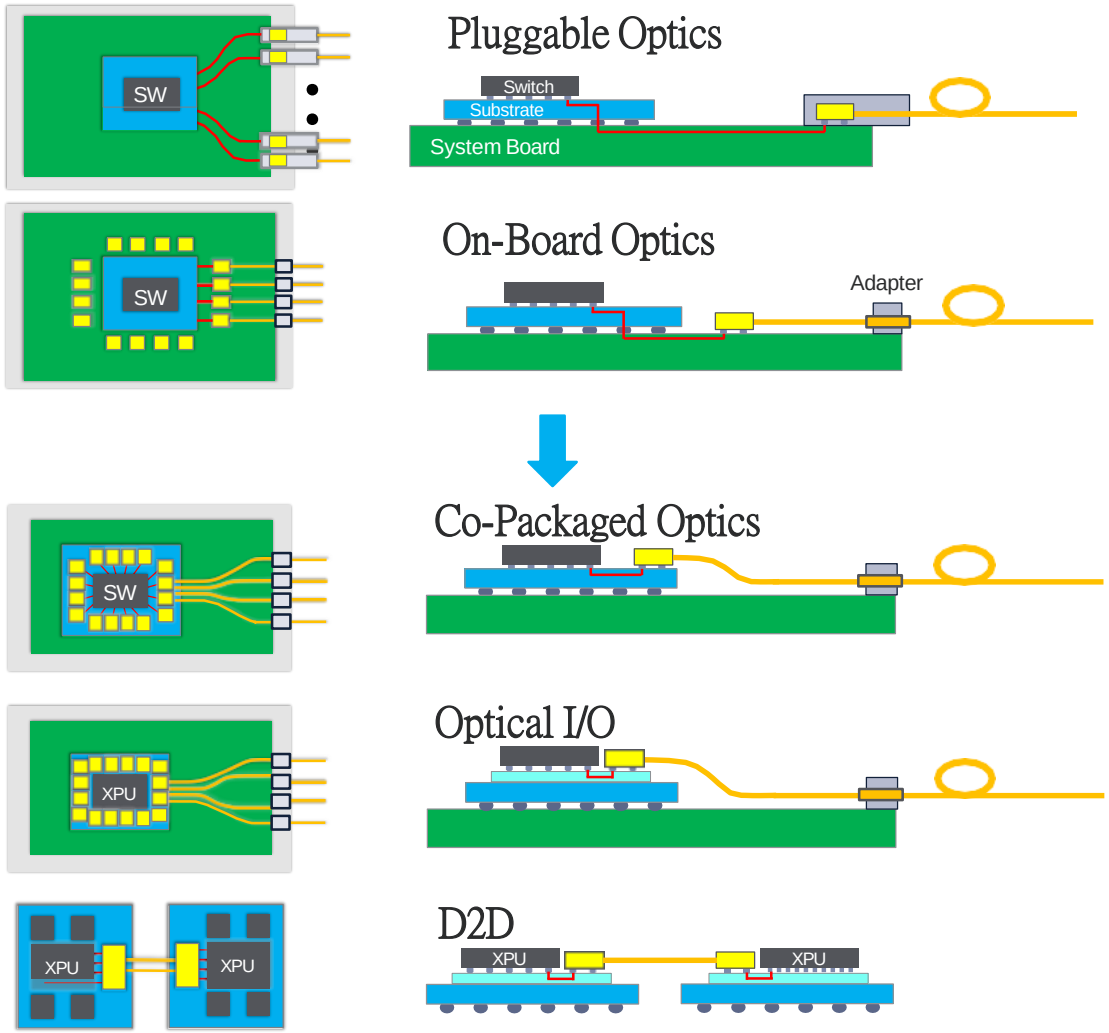
OSAT/CM
Optics
ASSY & Test

OSAT/CM
Module/OBO/CPO
ASSY & Test

SiPh Optics Integration Trend

Optical Interconnect

Shorter electrical interconnect



Packaging / Speed / Energy Efficiency

QSFP/OSFP



- 100/400/800G/1.6Tbps
- Shoreline 5~70G/mm
- Efficiency 20~25pJ/bit

CPO & Optical I/O PKG Requirement

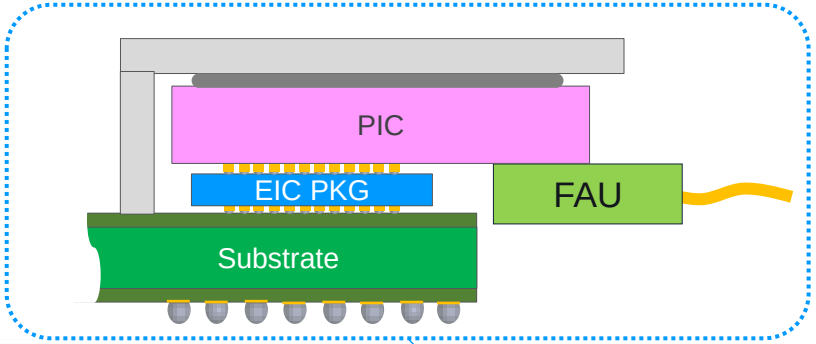
- High density PKG
- 3D packaging
- WG channel/pitch
- 224G SerDes
- WDM

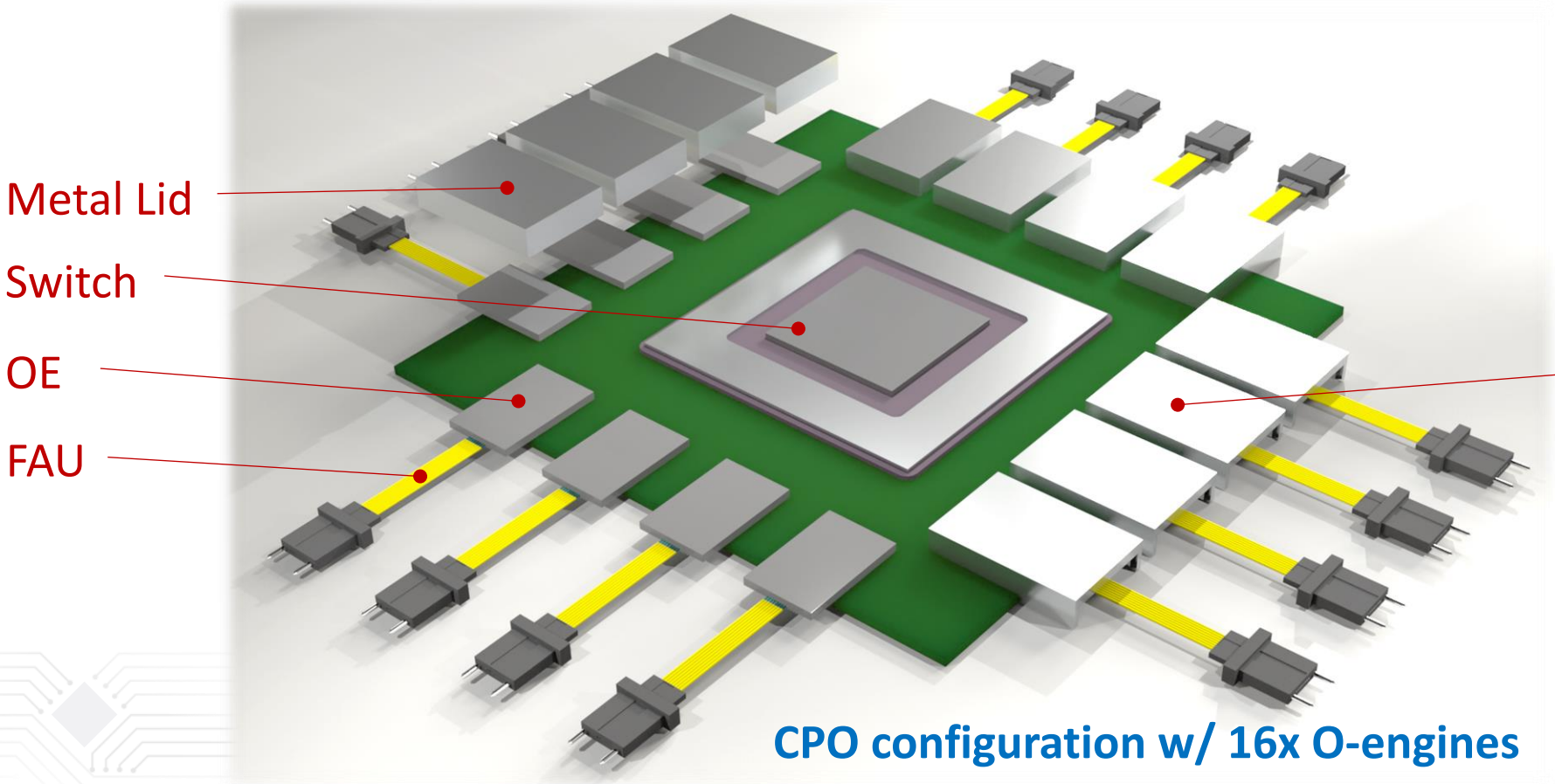
- 3.2/6.4/12.8Tbps
- Shoreline 200~800G/mm
- Efficiency ~5pJ/bit

- 4~32Tbps
- Shoreline 500G~2T/mm
- Efficiency <3pJ/bit
- Lower latency

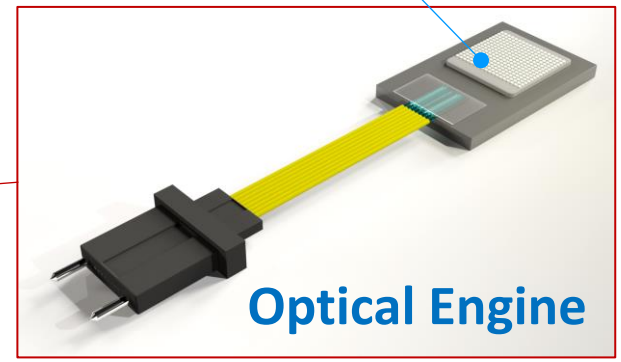
FOPOP Optical Engine in CPO

- **Optical Engine PKG Structure**
  - 3 essential blocks of O-engine: **PIC + EIC FO PKG + FAU**
  - EIC PKG: Double sided FO PKG
  - FAU: Detachable & Fixed approaches
  - LD: ELS



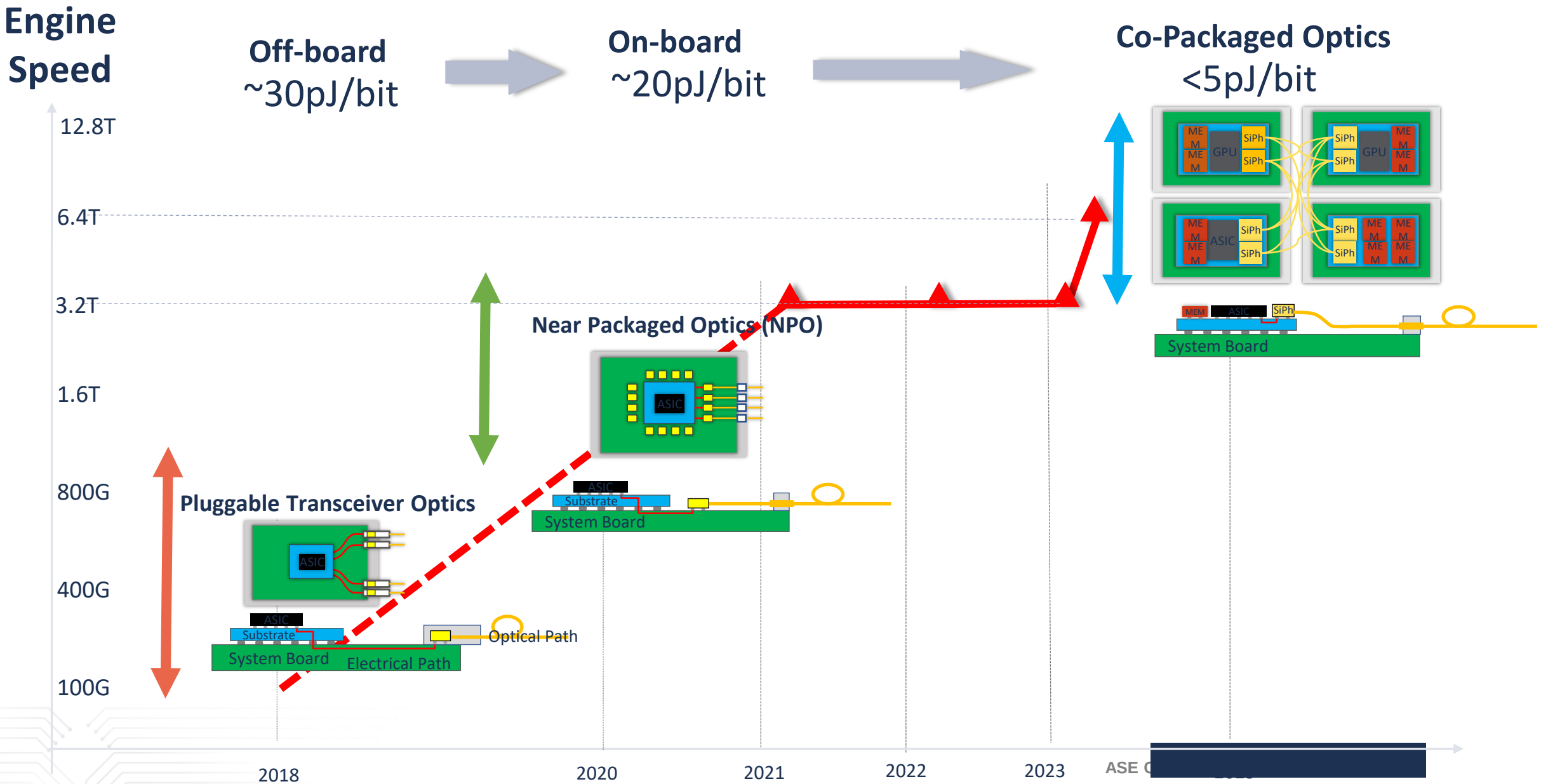


CPO configuration w/ 16x O-engines

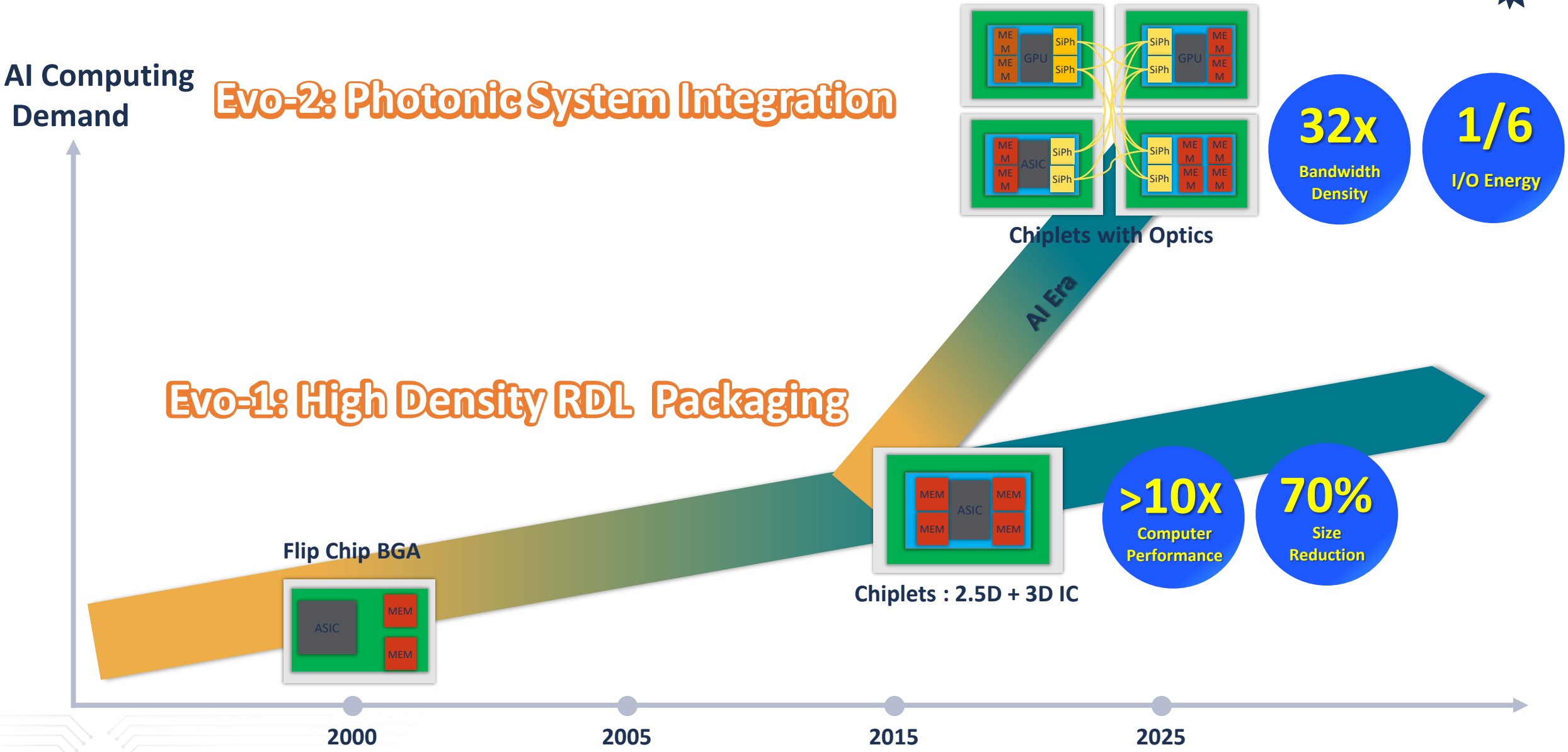


Si Photonics Engine Migrating Locations

Demo in OFC with Adv. SiP

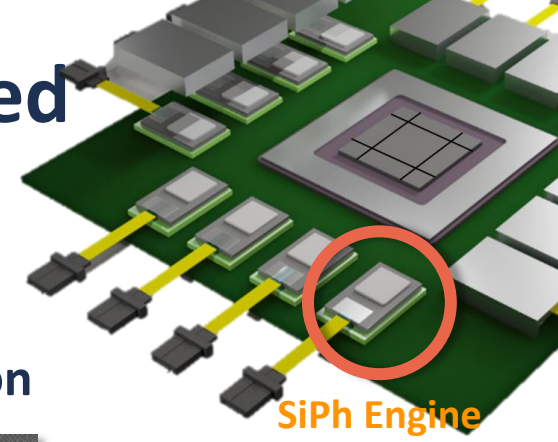


Packaging to System Evolution for AI Demand

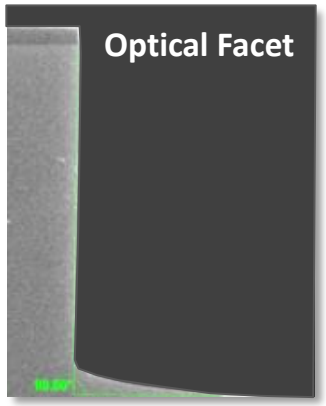


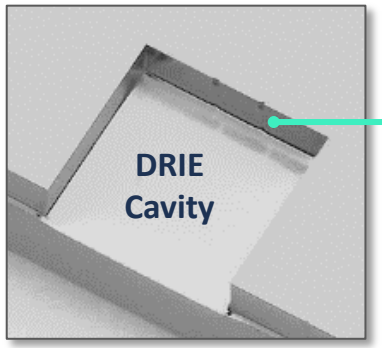
ASE Confidential / Security-B

SiPh Engine Toolbox for Optical Alignment & High Speed

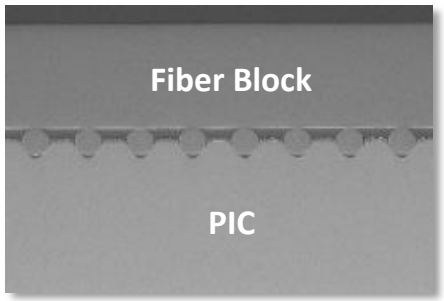


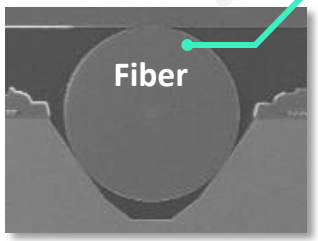
Optical Coupling Facet



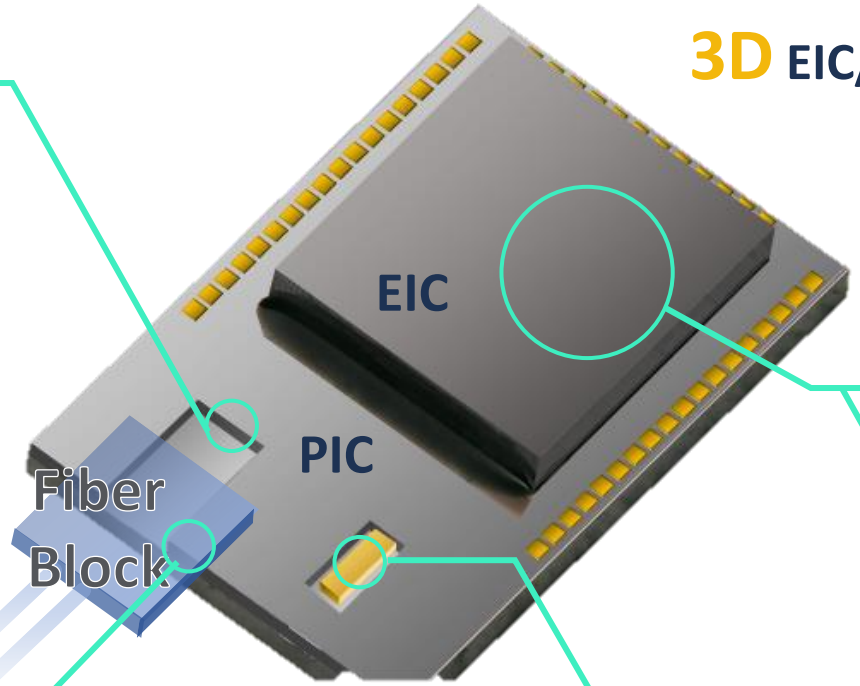


Enable 5x UPH of Fiber ASSY

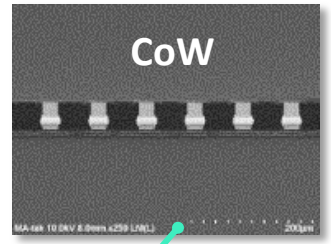


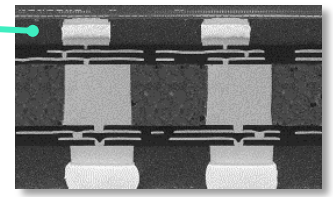


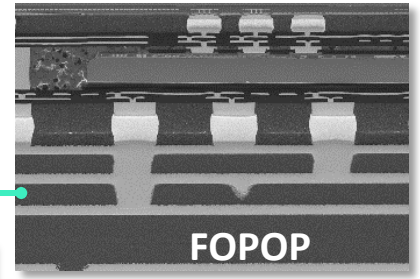
Passive fiber attachment

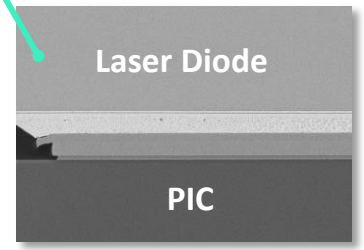


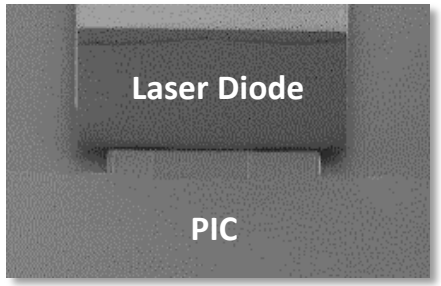
3D EIC/PIC Integration







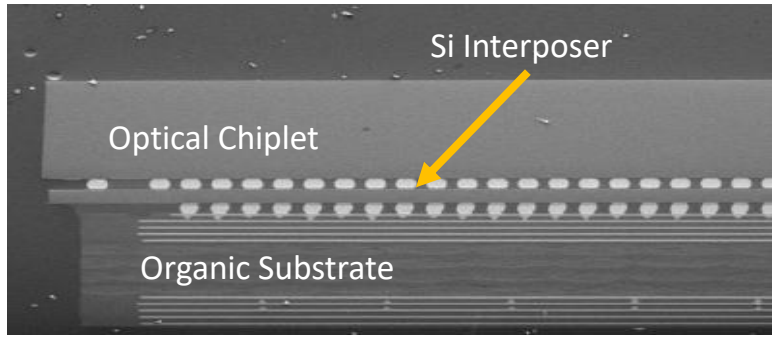




Laser Diode Integration ⟷ Sub-µm accuracy

Optics Integration – 3D Si Photonics

Optical Chiplet



Feature

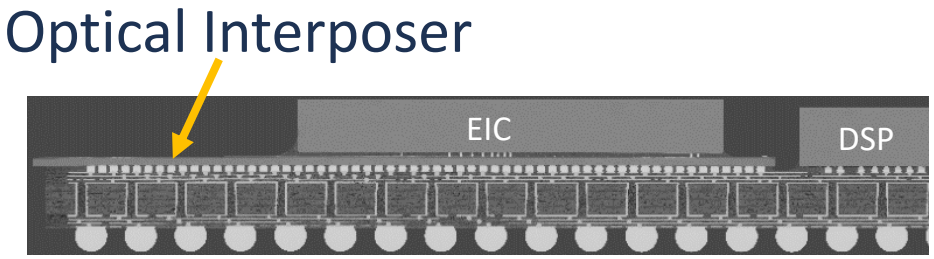
EIC & PIC Integrated on Si interposer

Application

Optical transceiver

Optical I/O for HPC

Optical Interposer



Feature

EIC on PIC (optical interposer)

Application

Optical Engine for Network Switch

Thank you